SMART SHOPPING ASSISTANT
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Project specifications

Project Overview

The purpose of this project is the implementation of an autonomous “Agentic commerce” platform, where users can make product requests (text and/or image), after which an AI agent autonomously searches through various online stores, calculates the best value and executes the final checkout process using single-use virtual cards.

Key features:

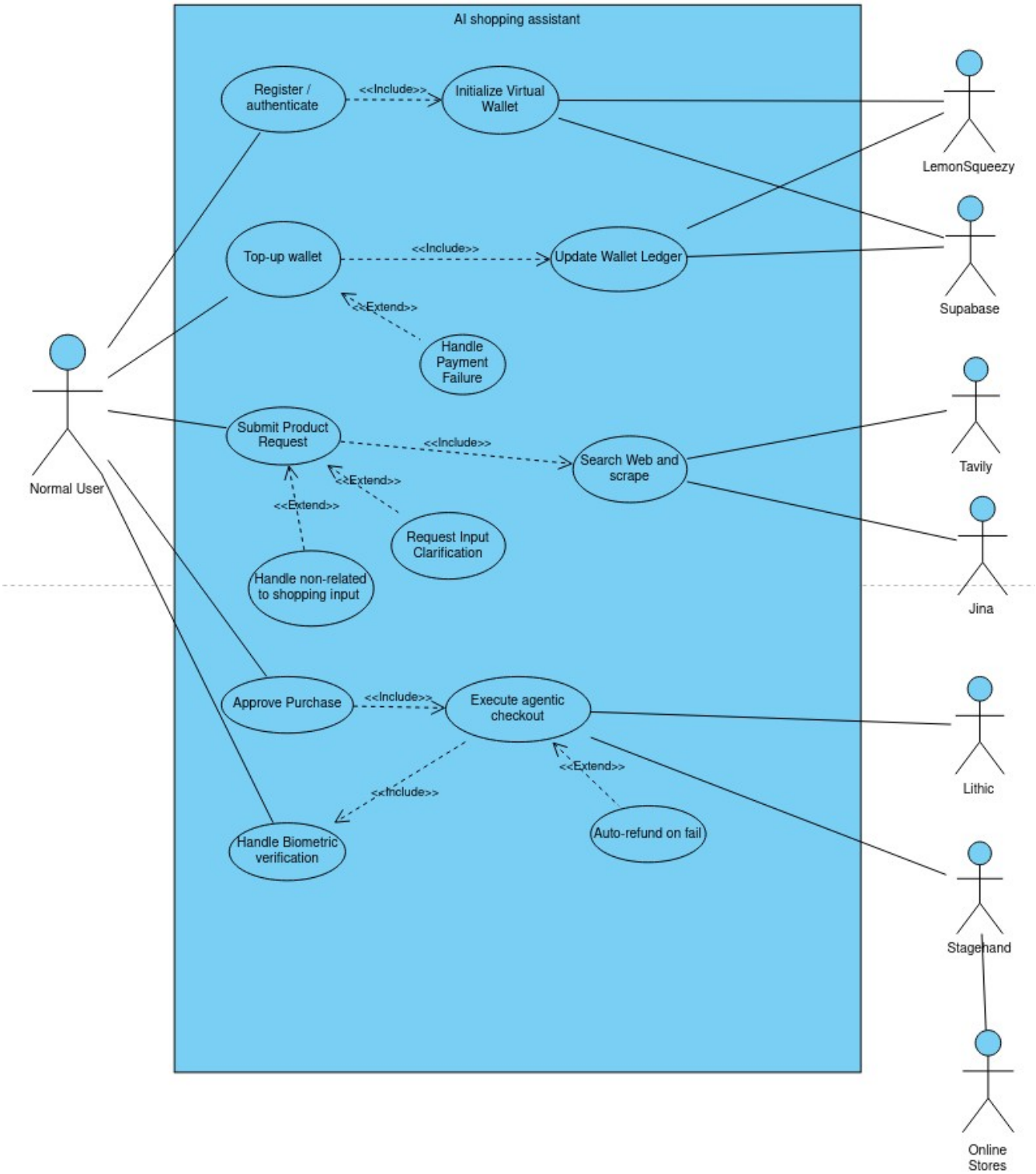
- Text and Image recognition: the AI agent must be able to recognize both text and images and interpret them
- Value scoring: the app always chooses the best product by calculating a value score from 1 to 100 depending on:
  - Cost efficiency: based on the base price, shipping cost and market delta
  - Product quality confidence: based on average rating and review volume
  - Logistics and convenience: delivery time and in-stock status
  - Trust and risk mitigation: Check if the product is sold directly by the provider or a 3rd party
- Agentic search: Uses Tavily API to find products across the web without being blocked and Jina to retrieve the content of the opened tab and turn it into a Markdown file
- Automated checkout: The user has to choose between the 3 best options, then after he chooses an option, the app automates the checkout process, creating a virtual card and then autocompletes with the virtual card details.
- Master-Ledger architecture: Whenever an user wants to buy something, a virtual card is created, and after the purchase, it is destructed
- Additional verification: Whenever an user buys something, he has to confirm by passkey

How it works:

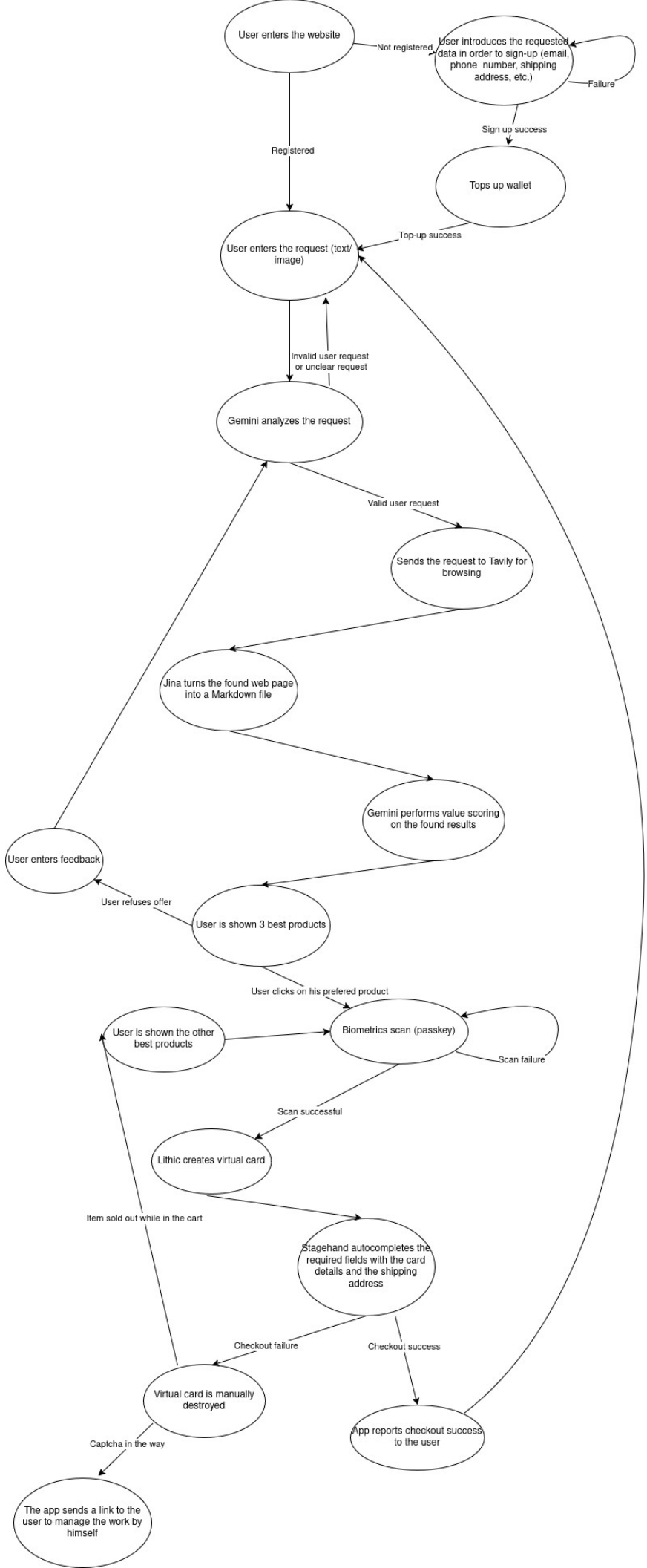
- Whenever an user creates an account, he has to introduce his phone number and email address, and a balance is created for him in the database.
- User types the request and optionally adds some pictures
- Gemini processes the user input

- It sends the request to Tavily to identify the highest quality products
- Jina converts the content of the web pages into readable Markdown files
- Gemini applies the value scoring function on all the products and displays the top 3 recommendations to the user
- The user selects one of the 3 items
- User verifies himself by providing a passkey
- Lithic creates a virtual card
- Stagehand autocompletes with the requested payment details (card details, shipping address, etc.)
- If checkout is successful, Lithic destroys the virtual card, otherwise it is destroyed manually and the user is shown the other best products in case of an out of stock situation, or in case of a captcha failure, it is provided a link to the respective website to manage the payment process by himself

Use case diagram:



State diagram:



Programming languages and API's:

- **Frontend**
  - Next.js, TypeScript, Tailwind CSS, Lucide React

- **Backend**
  - Python 3.12+
  - FastAPI
  - Pydantic (for data validation)

- **Databases**
  - Supabase (PostgreSQL)
  - SQL
  - Supabase Auth (for user login)

- **API's**
  - Lithic API - for generating virtual cards in the ledger architecture
  - LemonSqueezy API
  - Tavily API - provides AI-optimized web results
  - Jina Reader API – Converts website URL's into clean Markdown

- **AI tools**
  - Gemini 1.5 Flash – the core LLM for request recognition, image processing and value scoring
  - Stagehand – automates only the checkout part of the process
  - Model Context Protocol (MCP)

- **Deployment**
  - Vercel for frontend
  - Railway for backend
  - Supabase

Financial Projections (Estimated Operating Costs)

MVP Phase (0 to ~100 Users)

- **Fixed Infrastructure:** ~$5.00 / month (Vercel Free, Supabase Free, Railway Developer Plan).
- **Variable API Cost:** ~$0.075 USD (~0.35 RON) per successful agentic checkout.

Large Scale Phase (1,000+ Active Users)

- **Fixed Infrastructure:** ~$70.00 / month (Vercel Pro, Supabase Pro, Railway Scaled RAM).
- **Variable API Cost:** ~$0.075 USD (~0.35 RON) per successful agentic checkout.